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power_charge

File: power_charge.kicad_sch

battery_supply

File: battery_supply.kicad_sch

mcu

File: mcu.kicad_sch

sensing

File: sensing.kicad_sch

force

File: force.kicad_sch

stage_position

File: stage_position.kicad_sch

actuator_x

File: actuator_x.kicad_sch

actuator_y

File: actuator_y.kicad_sch

storage_comm

File: storage_comm.kicad_sch

user_interface

File: user_interface.kicad_sch

fault_handling

File: fault_handling.kicad_sch

Active stabilisation pen – research electronics Rev A. PROPOSED DESIGN: not built, not measured.
Every part marked VERIFY or SELECT in its fields must be checked against the named source before layout.
Power tree: USB–C –> MCP73831 –> cell (BQ29700 + 2 NMOS) –> TLV75530 (+3V0, +3V0A) and TPS22917 (VM0T, gated).
Calculations: electronics/calcs/drive_sense.py –> results/electronics/; ngspice drive stage: electronics/spice/.

RESEARCH ELECTRONICS – DEVELOPMENT DRAFT – NOT RELEASED FOR FABRICATION

Active stabilisation pen programme

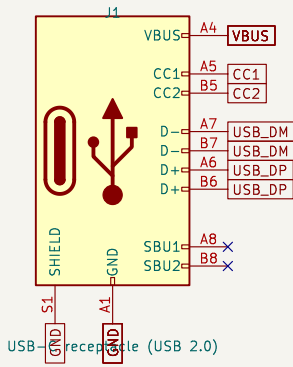
Sheet: /
File: pen_research.kicad_sch

Title: Active stabilisation pen – research electronics Rev A

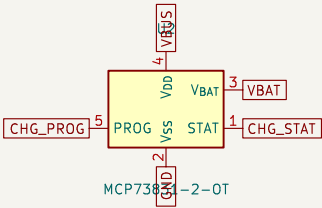
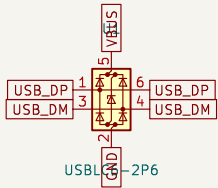
Size: A3Date: 2026–09–27Rev: A

KiCad E.D.A. 8.0.9Id: 1/12

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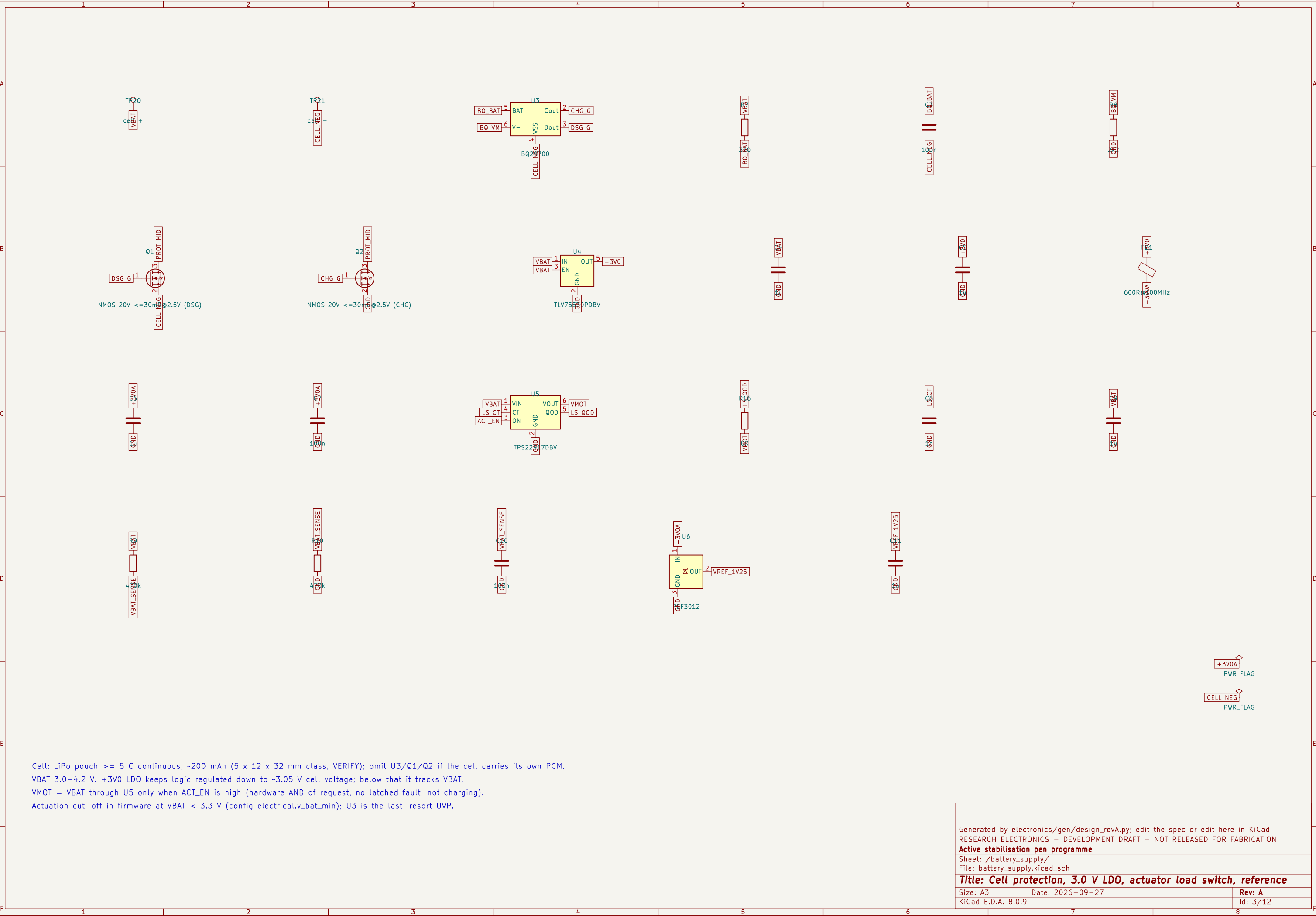
USB-receptacle (USB 2.0)



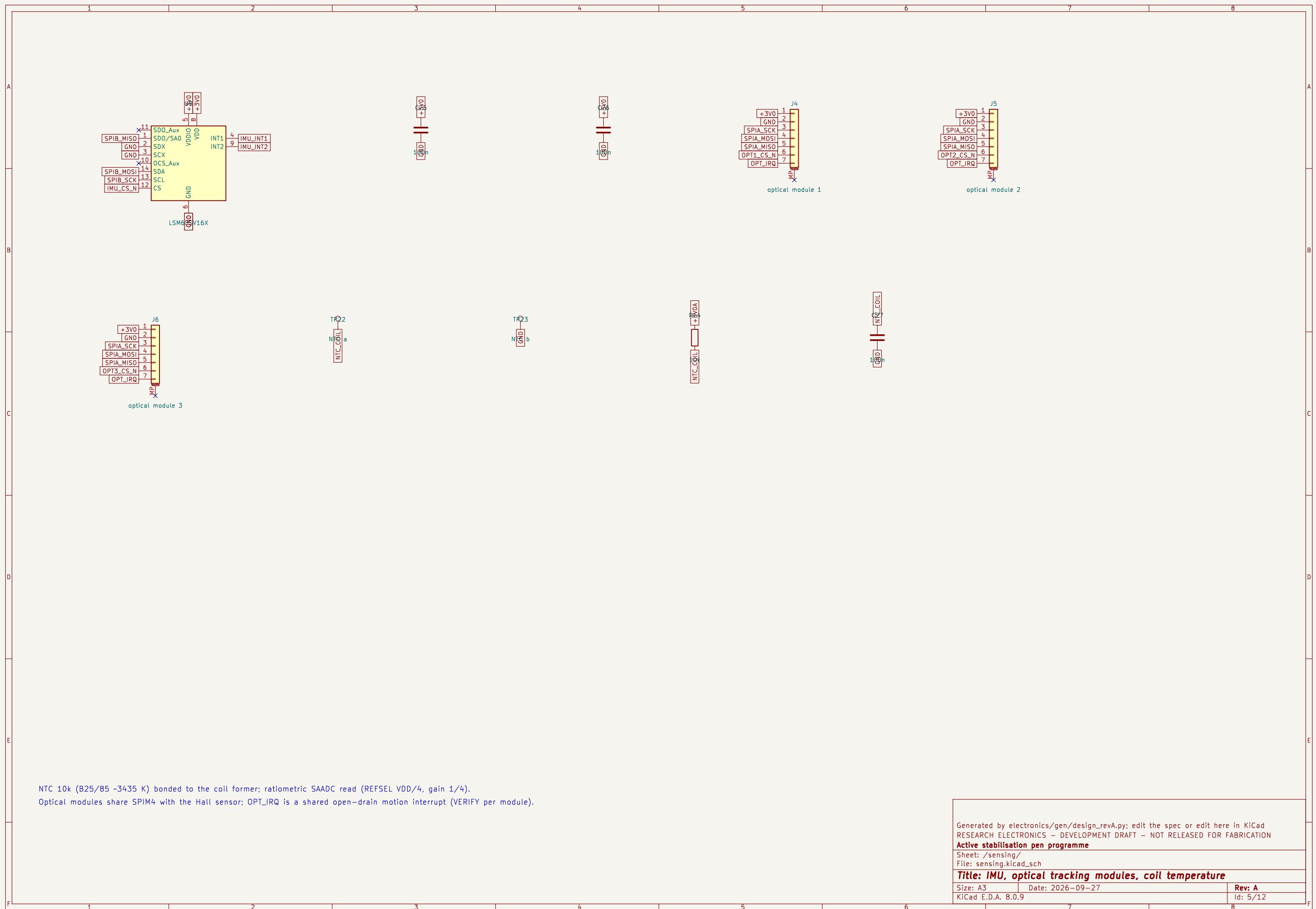
GND
PWR_FLAG

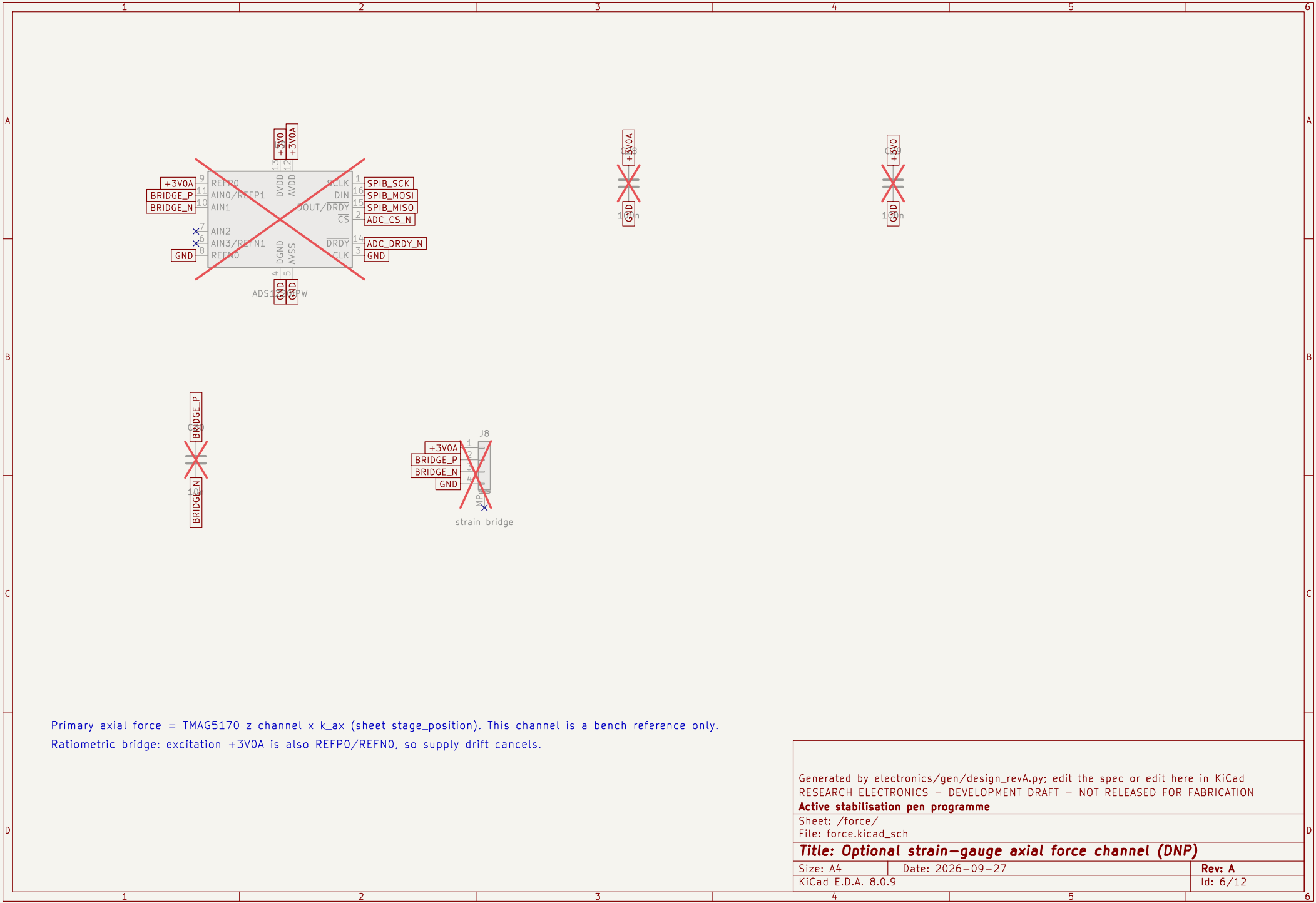
VBUS
PWR_FLAG

CHG_DET = VBUS/2 (2.37–2.62 V for 4.75–5.25 V VBUS): valid logic high at VDD 3.0 V, below VDD+0.3 V.
CHG_DET also drives the charging interlock (sheet fault_handling): actuation is disabled while VBUS is present.
Charge current 147 mA (R3): check against the selected cell's 0.5–1 C limit and the 43 C skin limit.



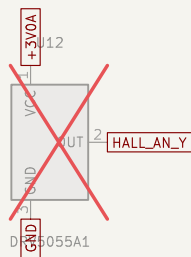
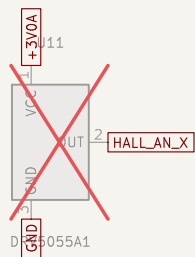
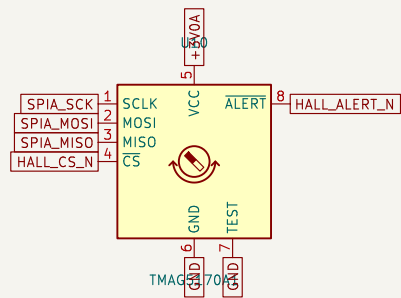






Primary axial force = TMAG5170 z channel x k_ax (sheet stage_position). This channel is a bench reference only.
Ratiometric bridge: excitation +3V0A is also REFPO/REFNO, so supply drift cancels.

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RESEARCH ELECTRONICS – DEVELOPMENT DRAFT – NOT RELEASED FOR FABRICATION		
Active stabilisation pen programme		
Sheet: /force/		
File: force.kicad_sch		
Title: Optional strain–gauge axial force channel (DNP)		
Size: A4	Date: 2026–09–27	Rev: A
KiCad E.D.A. 8.0.9		Id: 6/12



TMA65170 at the lever tail reads a sense magnet on the carrier: Bx, By → stage angles; Bz → axial slide s → $F_{ax} = F_{pre} + k_{ax} s$.
Cross-axis and actuator-field coupling calibrated on the bench (EXP-B04); sample mid-PWM to avoid coil-field ripple.

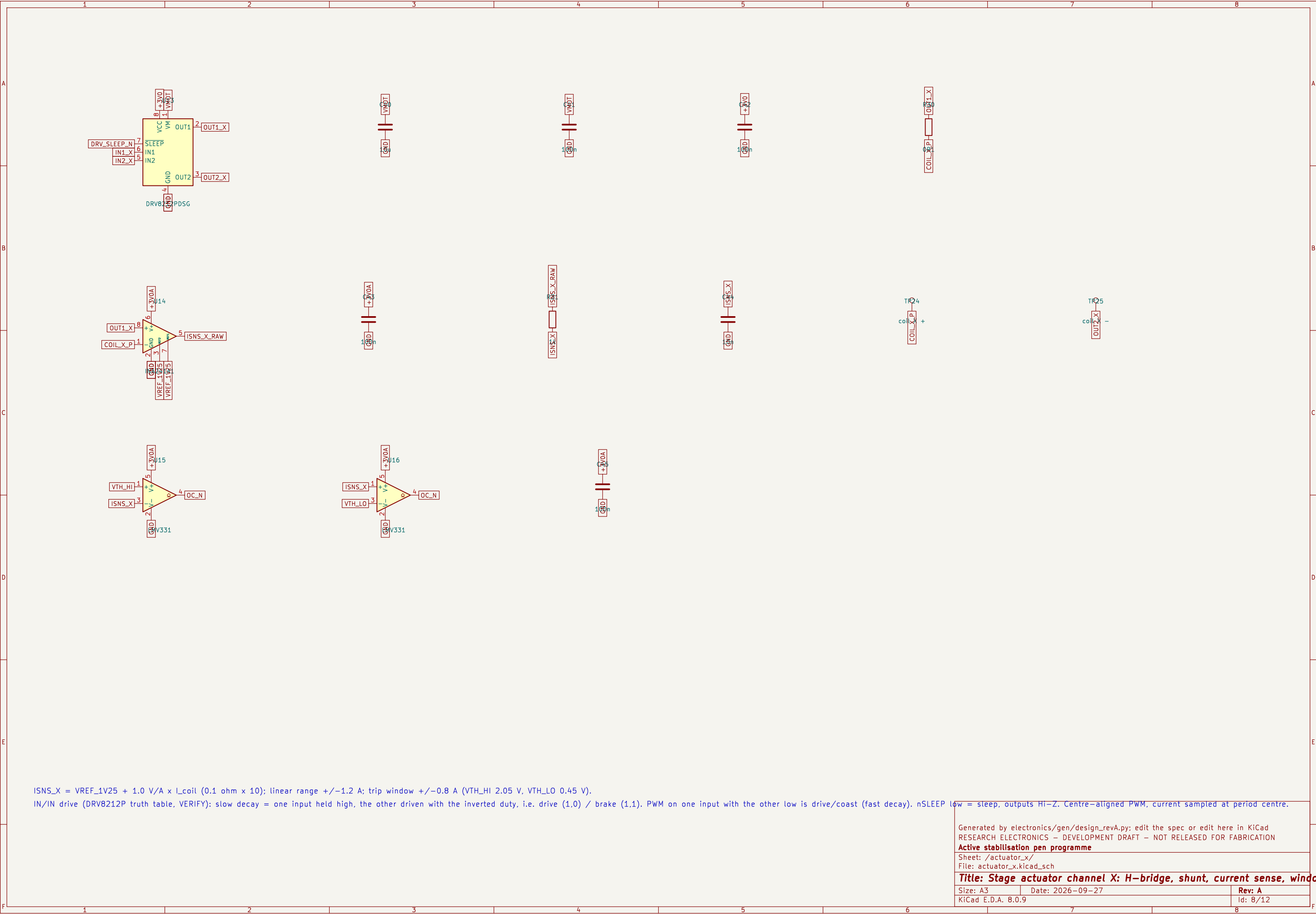
Generated by electronics/gen/design_revA.py; edit the spec or edit here in KiCad
RESEARCH ELECTRONICS – DEVELOPMENT DRAFT – NOT RELEASED FOR FABRICATION

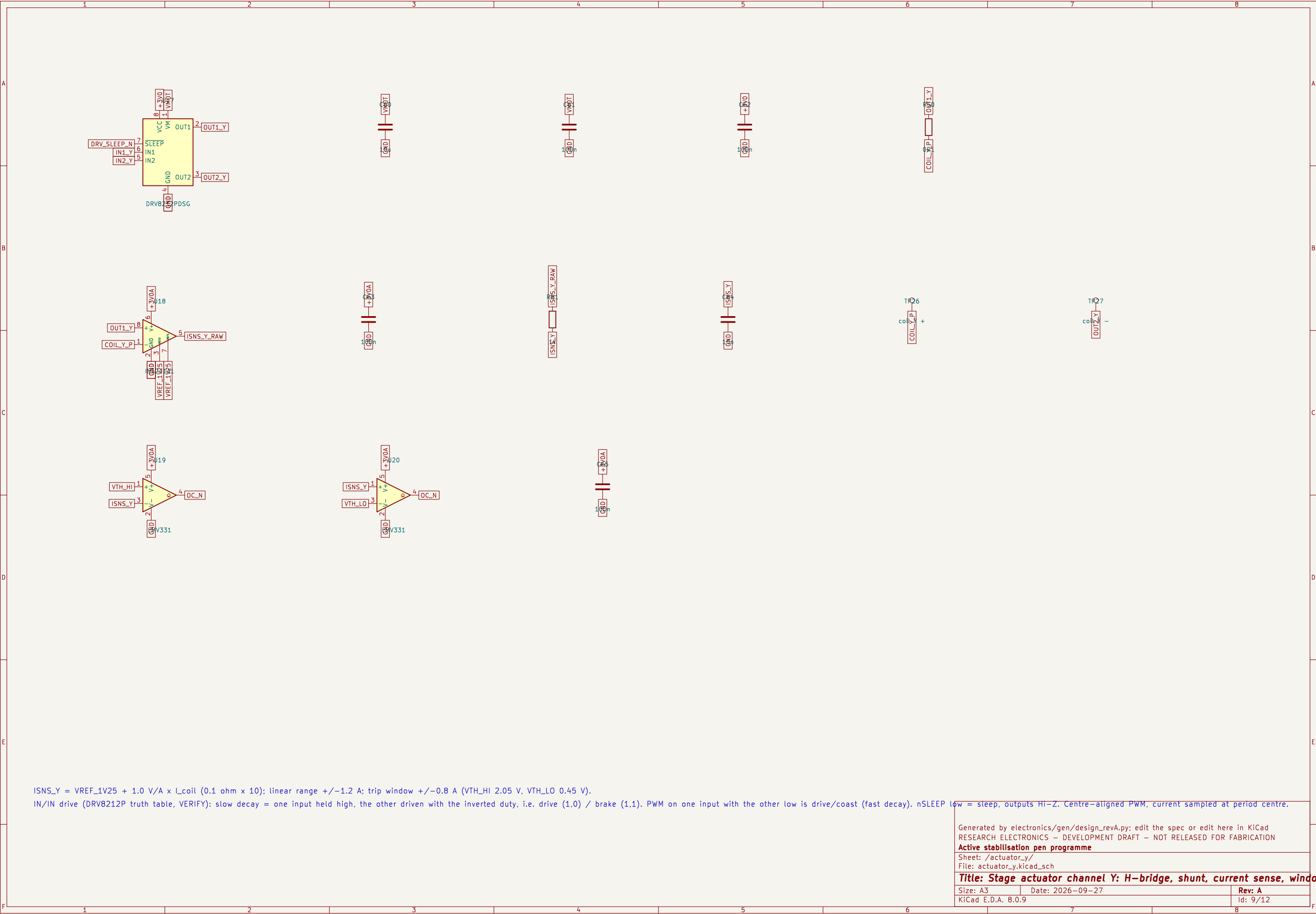
Active stabilisation pen programme

Sheet: /stage_position/
File: stage_position.kicad_sch

Title: Stage angle and axial slide sensing (3-axis Hall)

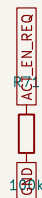
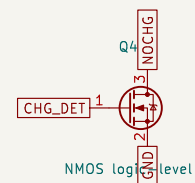
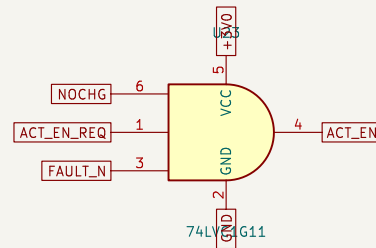
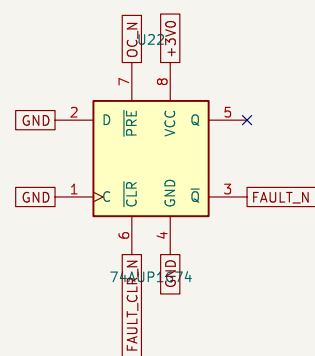
Size: A4	Date: 2026-09-27	Rev: A
KiCad E.D.A. 8.0.9	Id: 7/12	







ERM: coin vibration motor on flying leads (TP28/TP29); limit PWM duty to its rated voltage.



ACT_EN = ACT_EN_REQ AND FAULT_N AND NOCHG. Any comparator trip sets the latch (FAULT_N low) and opens VMOT within ~2 μ s + switch turn-off.

Latch clear sequence (firmware): ACT_EN_REQ low -> pulse FAULT_CLR_N low 1 us -> read FAULT_N high -> ACT_EN_REQ high.

Independent watchdog: MCU reset releases ACT_EN_REQ (R71 pull-down) and DRV_SLEEP_N (driver internal pull-down), so VMOT opens.

Thresholds from +3V0A: VTH_HI = 2.048 V (+0.80 A), VTH_LO = 0.455 V (−0.80 A) about VREF 1.25 V.